Example 8.2

Calculate the pH of 0.001M HCl solution

$$\underset{0.001\ M}{HCl} \xrightleftharpoons{H_2O} \underset{0.001\ M}{H_3O^+} + \underset{0.001M}{Cl^-}$$

H_3O^+ from the auto ionisation of H_2O $(10^{-7}M)$ is negligible when compared to the H_3O^+ from 10^{-3}M HCl.

Hence $[H_3O^+] = 0.001$ mol dm^{-3}

$$pH = -\log_{10}[H_3O^+]$$
$$= -\log_{10}(0.001)$$
$$= -\log_{10}(10^{-3}) = 3$$

Note: If the concentration of the acid or base is less than 10^{-6} M, the concentration of H_3O^+ produced due to the auto ionisation of water cannot be negleted and in such cases

$[H_3O^+] = 10^{-7}$ (from water) + $[H_3O^+]$ (from the acid)

similarly, $[OH^-] = 10^{-7}$M (from water) + $[OH^-]$ (from the base)

Example 8.3

Calculate pH of 10^{-7} M HCl

If we do not consider $[H_3O^+]$ from the ionisation of H_2O,

then $[H_3O^+] = [HCl] = 10^{-7}$M

i.e., pH = 7, which is a pH of a neutral solution. We know that HCl solution is acidic whatever may be the concentration of HCl i.e, the pH value should be less than 7. In this case the concentration of the acid is very low $(10^{-7}M)$ Hence, the H_3O^+ $(10^{-7}M)$ formed due to the auto ionisation of water cannot be neglected.

so, in this case we should consider $[H_3O^+]$ from ionisation of H_2O

$$[H_3O^+] = 10^{-7} \text{ (from HCl)} + 10^{-7} \text{ (from water)}$$
$$= 10^{-7}\ (1+1) \quad = 2\times10^{-7}$$
$$pH = -\log_{10}[H_3O^+]$$
$$= -\log_{10}(2\times10^{-7}) = -\left[\log 2 + \log 10^{-7}\right]$$
$$= -\log 2 - (-7).\log_{10} 10$$
$$= 7 - \log 2$$
$$= 7 - 0.3010 = 6.6990$$
$$= 6.70$$

XII U8-Ionic equilibrium.indd 11 16-12-2022 19:33:46